



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO NS TRADERS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

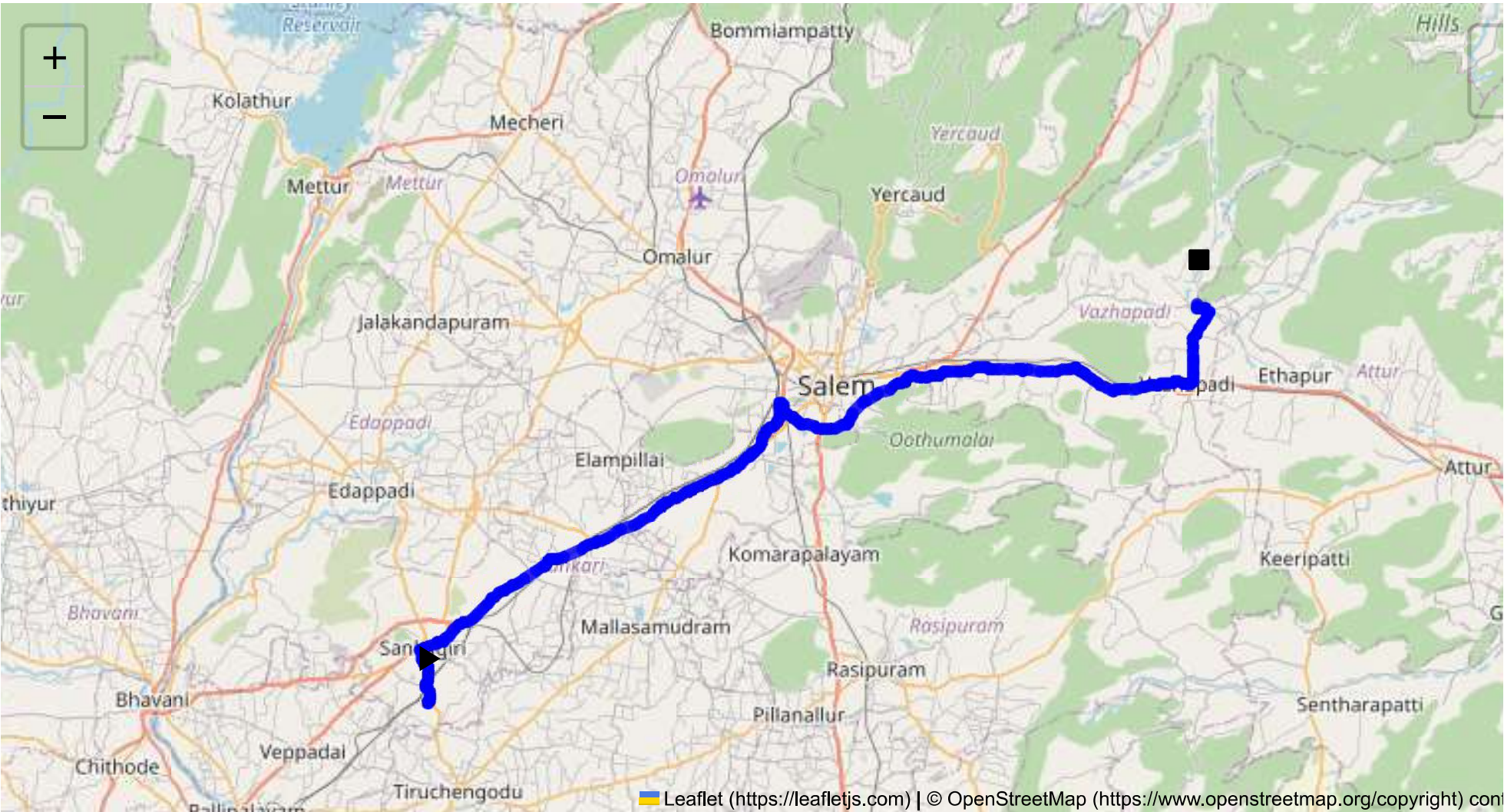
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 80.36 km**  
**Estimated Duration: 1.7 hours**  
**Adjusted Duration (Heavy Vehicle): 2.1 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.709613, 78.408576)**



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 4397/5, Kurichi, Belur spans approximately 80.36 kilometers. The primary route often taken includes traveling along NH44, which is a major highway that connects several key areas and offers a relatively direct path between these two points.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** Tamil Nadu generally has a tropical climate, with hot summers (March to June), a wet monsoon season (June to September), and a cooler, drier winter (November to February).
- **Weather-Related Hazards:** Monsoon rains can lead to heavy downpours, causing flooding and reduced visibility. The region may also experience thunderstorms during this period, which could affect driving conditions.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, the morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) are peak hours with possible congestion, especially near major towns and urban centers.
- **Congestion-Prone Areas:** Areas near city centers such as Salem and intersections with significant urban traffic are most prone to congestion.

## 4. Assessment of Road Quality and Infrastructure

- **NH44:** Generally maintained well, with adequate signage and road markings.
- **Secondary Roads:** As you approach Belur, road quality can vary with potential for narrow lanes and less frequent maintenance.
- **Infrastructure:** Be cautious of potential construction zones and agricultural machinery on secondary roads.

## 5. Suggestions for Alternative Routes for Emergencies

In case of major roadblocks or emergencies, an alternative route could involve using SH84 via Omalur and transitioning back onto NH44, although this may extend the travel time.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Permits are required for transporting hazardous materials, especially in populated areas. Restrictions also exist for transportation through certain sensitive zones during peak hours.

## 7. Overview of Historical Incidents

There have been occasional incidents involving heavy vehicles on NH44 related to vehicular failure or adverse weather conditions. These incidents generally result in temporary traffic stagnation and highlight the need for regular vehicle maintenance.

## 8. Environmental Considerations and Sensitive Areas

- Areas of dense population and marketplaces should be approached with caution due to the presence of pedestrians and cross-traffic.

- Proximity to agricultural land poses a risk of animal crossings.

## 9. Analysis of Communication Coverage

- **Major Highways:** Generally good mobile and internet coverage.
- **Rural Areas:** Potential dead zones may exist on secondary roads away from major urban centers.

## 10. Estimated Emergency Response Times

- **NH44 Segments:** Quick response times due to proximity to larger towns.
- **Rural Areas Near Belur:** Slower response times due to distance from major emergency services.

## 11. Overall Summary of Risk Assessment

The route, while generally safe, presents common risks associated with heavy traffic, variable road conditions, and weather impacts, particularly during the monsoon. Preparing for potential delays and ensuring compliance with local regulations are essential for safe transport. Regular communication with monitoring stations is recommended to navigate dead zones and ensure timely responses to emergencies. The infrastructure support along the highway is robust, but vigilance is required in more remote areas.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.63303, 78.12713 | 15 KM/Hr    |
| 1  | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.45000, 77.87213 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 9  | Turn       | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 10 | Turn       | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.61736, 78.10588 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.61753, 78.10608 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 14 | Turn      | Medium     | 11.64172, 78.11817 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.64216, 78.11875 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.64195, 78.12129 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.64162, 78.12160 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.64120, 78.12168 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.63948, 78.12081 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.65459, 78.37964 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.65472, 78.37991 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.65484, 78.38000 | 30 KM/Hr    |
| 23 | Turn      | Medium     | 11.65602, 78.38711 | 30 KM/Hr    |
| 24 | Turn      | High       | 11.65716, 78.40570 | 15 KM/Hr    |
| 25 | Turn      | Medium     | 11.65743, 78.40584 | 30 KM/Hr    |
| 26 | Turn      | High       | 11.70462, 78.41562 | 15 KM/Hr    |
| 27 | Turn      | High       | 11.70565, 78.41336 | 15 KM/Hr    |
| 28 | Turn      | Medium     | 11.70703, 78.41360 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 11.70740, 78.41345 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 11.70789, 78.41265 | 30 KM/Hr    |
| 31 | Turn      | High       | 11.70789, 78.40774 | 15 KM/Hr    |

Emergency Locations

|   | type     | name                    | coordinates            | speed_limit | risk_level |
|---|----------|-------------------------|------------------------|-------------|------------|
| 0 | hospital | Kathir Eye Hospital     | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 1 | police   | Police Quarters         | 11.4741243, 77.8673495 | 30 km/h     | Medium     |
| 2 | hospital | Thilagam Hospital       | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 3 | clinic   | Meyyazhagan Clinic      | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 4 | hospital | Dr. Jaganathan Hospital | 11.4752717, 77.8709312 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 6  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 8  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 9  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 11 | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 12 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 13 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 14 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 17 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 21 | hospital | Priyam Speciality Hospital                                      | 11.63446,<br>78.124388    | 30 km/h     | Medium     |
| 22 | hospital | Universal Cancer Hospital                                       | 11.6336739,<br>78.1267767 | 30 km/h     | Medium     |
| 24 | hospital | AVM Hospital                                                    | 11.6264369,<br>78.1394204 | 30 km/h     | Medium     |
| 25 | hospital | Dharan Hospital                                                 | 11.6252302,<br>78.1451753 | 30 km/h     | Medium     |
| 26 | hospital | Udhayam Hospital                                                | 11.625819,<br>78.145958   | 30 km/h     | Medium     |
| 27 | pharmacy | மருந்தகம்                                                       | 11.6251501,<br>78.1470098 | 30 km/h     | Medium     |
| 28 | hospital | மருத்துவமனை நிலம்                                               | 11.625344,<br>78.146058   | 30 km/h     | Medium     |
| 29 | hospital | Manima Clinic                                                   | 11.623248,<br>78.1465721  | 30 km/h     | Medium     |
| 30 | hospital | Rps Hospital                                                    | 11.6260528,<br>78.1474921 | 30 km/h     | Medium     |
| 31 | hospital | Akitha Hospital                                                 | 11.6256027,<br>78.1489875 | 30 km/h     | Medium     |
| 32 | hospital | Deepam Speciality Hospital                                      | 11.6269734,<br>78.1476357 | 30 km/h     | Medium     |
| 36 | hospital | Jaaidev Hospital                                                | 11.6655205,<br>78.2634736 | 30 km/h     | Medium     |

|    | type     | name                          | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------|---------------------------|-------------|------------|
| 38 | hospital | Niharika Hospital             | 11.6565971,<br>78.3937356 | 30 km/h     | Medium     |
| 39 | hospital | Malar Hospital                | 11.6567854,<br>78.3928345 | 30 km/h     | Medium     |
| 40 | hospital | Sri Bala Hospital             | 11.6566667,<br>78.4010889 | 30 km/h     | Medium     |
| 41 | clinic   | Sibi Physiotherapy Clinic     | 11.6561305,<br>78.3994263 | 30 km/h     | Medium     |
| 42 | hospital | Saraswathi Hospital, Valapady | 11.6563505,<br>78.4009062 | 30 km/h     | Medium     |
| 43 | hospital | Devaki Hospital               | 11.6564189,<br>78.402376  | 30 km/h     | Medium     |
| 44 | hospital | Sri Udaya Hospital            | 11.6560244,<br>78.4040862 | 30 km/h     | Medium     |
| 45 | hospital | Government Hospital, Valapady | 11.65706,<br>78.403773    | 30 km/h     | Medium     |
| 46 | hospital | Deepam Hospital               | 11.6576703,<br>78.4065155 | 30 km/h     | Medium     |

Crowded Spots

|    | type        | name                                                    | coordinates               | speed_limit | risk_level |
|----|-------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 5  | school      | Goverment Boys Higher Secondary School                  | 11.477659,<br>77.8659632  | 30 km/h     | Medium     |
| 7  | marketplace | Sunday Market                                           | 11.4796502,<br>77.8708458 | 30 km/h     | Medium     |
| 15 | university  | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 16 | school      | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 18 | school      | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 19 | school      | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 20 | school      | Diamond Rays Matric HR Sec School                       | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |



|    | type    | name                                       | coordinates               | speed_limit | risk_level |
|----|---------|--------------------------------------------|---------------------------|-------------|------------|
| 23 | school  | Sowdeshwari College                        | 11.6330643,<br>78.1276021 | 30 km/h     | Medium     |
| 33 | school  | Ganesh College                             | 11.6566575,<br>78.1991121 | 30 km/h     | Medium     |
| 34 | school  | Thangam ZEE public school                  | 11.6653353,<br>78.251641  | 30 km/h     | Medium     |
| 35 | college | AVS College od Arts and Science,<br>Salem. | 11.6642672,<br>78.2584134 | 30 km/h     | Medium     |
| 37 | college | Mahendra Engineering College               | 11.6672537,<br>78.2641303 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.63303, 78.12713





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.44900, 77.87406**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45000, 77.87213**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45309, 77.87325





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49149, 77.89601





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.61736, 78.10588





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.61753, 78.10608**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64172, 78.11817**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64216, 78.11875**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64195, 78.12129



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64162, 78.12160





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64120, 78.12168



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63948, 78.12081



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65459, 78.37964





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65472, 78.37991



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65484, 78.38000





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.65602, 78.38711**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.65716, 78.40570**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.65743, 78.40584**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70565, 78.41336

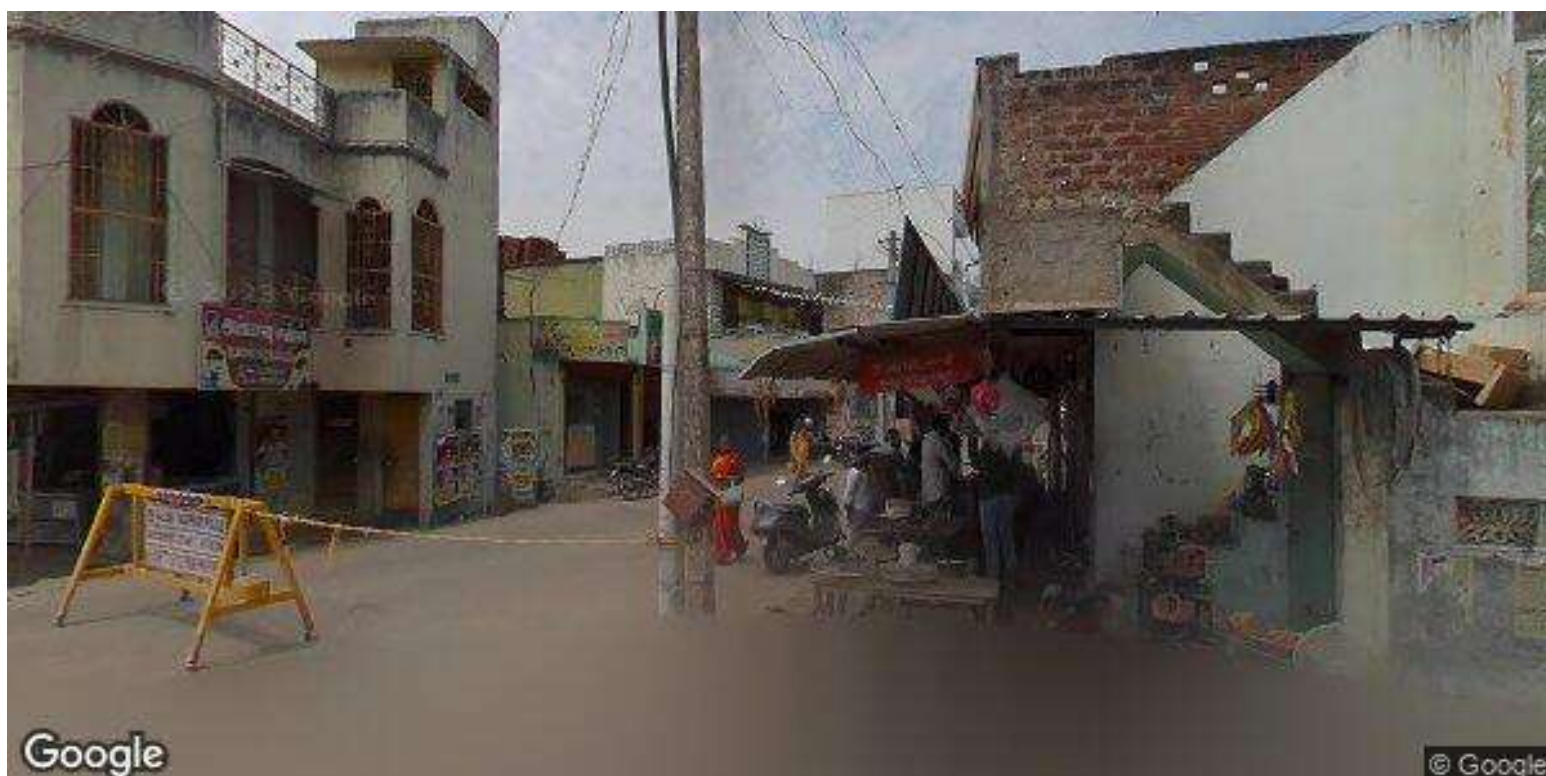


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.70703, 78.41360





Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70740, 78.41345



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70789, 78.41265



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70789, 78.40774



